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ASHRAE LEVEL I ENERGY AUDIT REPORT



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Abbreviations

A	Amperage
AC	Air Conditioning
Amps	Amperage
BMS	Building Management System
CHW	Chilled Water System
CF	Compact Fluorescent
CO ₂	Carbon Dioxide
COP	Coefficient of Performance ($KW_{cooling}/KW_{electrical}$)
EEM	Energy Efficiency Measures
ESCO	Energy Services Company
Fluo	Fluorescent
GHG	Green House Gases
Hal	Halogen
HVAC	Heating Ventilation Air Conditioning
IRR	Internal Rate of Return
KVA	kilo Volt Ampere
KW	Kilowatt
KWh	Kilowatt-hour
LED	Light Emitting Diode
m	Meter
m ²	Square meter
M3	Cubic meter
MDB	Main Distribution Board
MH	Metal Halide
MJ	Mega Joules
°C	Degree Celsius
°K	Degree Kelvin
°F	Degree Fahrenheit
PF	Power Factor
RT	Refrigerated Tons
TR	Transformer
UPS	Uninterruptible Power Supply
V	Volt
VAV	Variable Air Volume
VFD	Variable Frequency Drive
W	Watt
W m ² / °K	Watt Square meter per degree Kelvin

Conversion Factors

from	to	multiply by
meter	Feet	3.2808399
meter	Inches	39.370079
kg	Pounds	2.2046226
tons	Kg	1000
liter	Gallons (US beer)	0.264179
liter	Cubic feet	0.035315
kWh	MJ	3.6
kWh	BTU	3412
W/m ²	BTU/ft ²	0.317
kJ/kg K	BTU/lb °F	0.2388
W (heat flow rate)	BTU/h	3.412
W/m ² k	BTU/ft ² h °F	0.1761
kJ/kg	BTU/lb	0.4299
MJ/m ³	BTU/ft ³	26.84
Mm Hg (mercury)	Pa	133.332
Ft of water	kPa	2.98898
m ³ /kg	ft ³ /lb	16.02
m/s	ft/s	3.281

1. Introduction

Following an agreement with the Hotel Management, Energy Efficiency Group **EEG** has undertaken an energy audit of the Four Seasons Madrid Hotel, located in Spain (**abbreviated by FSM throughout this report**).

The completed energy audit's procedures are in between those listed between ASHRAE Level 1 and 2. A close cooperation was made with the facility department throughout the audit who showed great support and provided quality assistance.

The present report is divided in two main sections. The first section includes a general description of the utilities costs (energy baseline) and the operation of all existing equipment (energy balance). The second section of the report comprises a list of suggested energy efficiency measure with

1.1. Energy Efficiency Rationale

Simply put, Energy Efficiency is using less energy to accomplish the same amount of work needed. Ever since energy emerged as a top priority for policymakers, it has become a critical economic issue. In other words, unsustainable energy supply has real implications. Efficient buildings offer considerable opportunities to save money while reducing greenhouse gas emissions and taking leadership in sustainable development.

Within all sectors, energy costs may be a small percentage of turnovers, but reducing those costs can directly impact the profitability as they go straight to the bottom line, making businesses more competitive, especially with the rising of energy prices. The implementation of energy efficiency and sustainability measures increases levels of occupants' comfort as well as improving staff's general morale.

Energy audits include a review of all energy systems such as cooling, lighting, heating, ventilation, and building envelope. An analysis then serves the purpose of identifying where and how a facility uses energy and developing the set of Energy Efficiency Measures [EEMs] that could optimize the overall energy costs and which can range from small energy projects to major capital investments on operations and maintenance.

The audit focuses on a first phase on Actions with little or no cost which are improvements in current operations and practices that involve little or no cost of execution [LCNC]. Then it develops a road map for additional Energy efficiency measures that would cover all loads and infrastructure and will have different levels of CAPEX (Capital Expenditures) and Prioritization.

2. Facility General Description

2.1. Introduction

The Four Seasons Madrid Hotel (**abbreviated as *FSM* in this report**) is a five-star hotel that started operation in September 2020 and consists of 200 rooms, with 161 standard rooms and 39 suites over a total built up area of about 31,200m². There are also 22 private residences in the premises but are completely disconnected from the hotel's technical infrastructure. The building is aiming to be LEED certified and has been developed and constructed with a large number of energy efficiency and sustainability features.

Figure 1: FSM Facade (Left) and Main Lobby (Right)



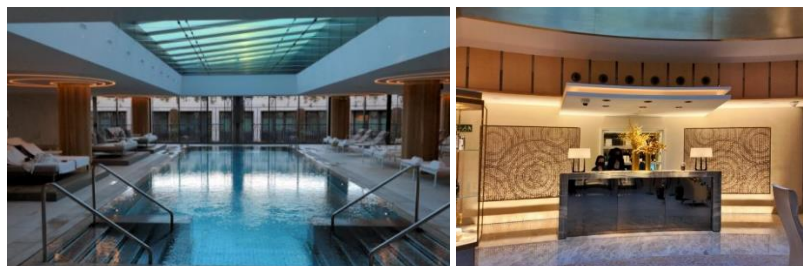
There are three F&B outlet in the hotel: Dani Restaurant on the last floor, and two lounges, El Patio and Isa, which opened recently in December 2021.

Figure 2: Isa Lounge (Left) and Dani Restaurant (Right)



FSM has a total of 8 floors, with the SPA located on the 5th floor, treatment rooms on the 4th, gym on the 6th, and an indoor pool on the 8th.

Figure 3: Indoor Pool (Left) and SPA Entrance (Right)



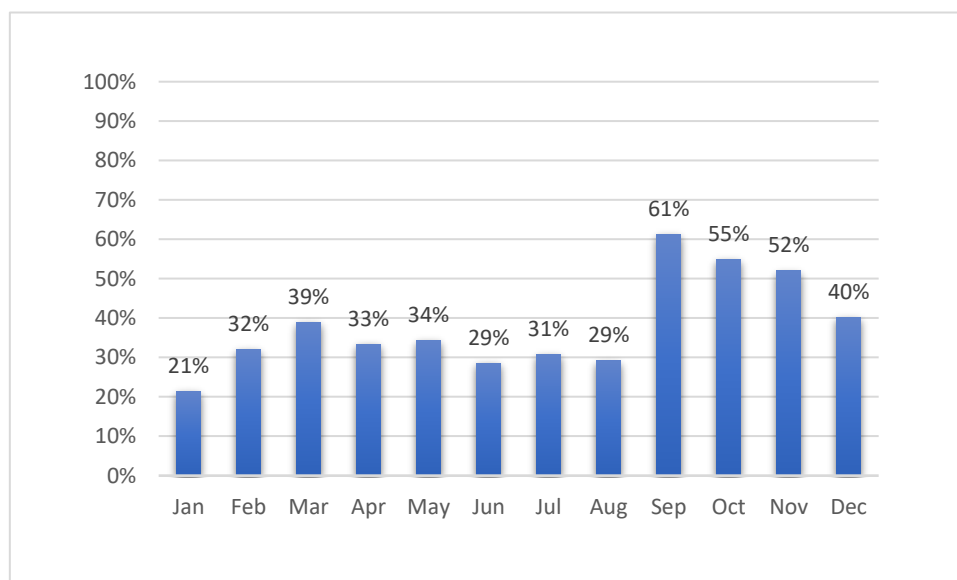
2.2. Occupancy

The following table depicts the occupancy rate variation of the hotel during the baseline year which was chosen to be 2021. As can be seen, the occupancy was fluctuating during the baseline period between a minimum of 21% in January and a maximum of 61% in December. The average occupancy rate throughout the baseline year is 38%.

Table 1: Monthly Occupancy Rate in FSM through the Baseline Period

Month	Occupancy	Availability	Occupancy Rate
Unit	Room Nights	Room Nights	%
Jan	1,323	6,200	21%
Feb	1,796	5,600	32%
Mar	2,414	6,200	39%
Apr	2,004	6,000	33%
May	2,130	6,200	34%
Jun	1,714	6,000	29%
Jul	1,914	6,200	31%
Aug	1,807	6,200	29%
Sep	3,675	6,000	61%
Oct	3,405	6,200	55%
Nov	3,120	6,000	52%
Dec	2,487	6,200	40%
TOTAL	27,789	73,000	38%
Maximum	3,675 [Sep]	6,200 [Jan]	61% [Sep]
Average	2,316	6,083	38%
Minimum	1,323 [Jan]	5,600 [Feb]	21% [Jan]

Figure 4: Monthly Occupancy Rate Variations



3. Utilities Accounting

The following sections detail the accounting numbers for the three utilities over the **baseline period from January 2021 until December 2021**. The annual consumption and related cost baseline are the aggregated consumption and associated costs for FSM throughout the baseline period. This energy baseline serves as a benchmark consumption to assess the current energy performance along with the calculation of all the proposed Energy Efficiency Measures. The present energy audit and the related energy simulation tackle the full resort space including all the restaurants, lounges, kitchens, and SPA.

3.1. Electricity

3.1.1. Electricity Overall Consumption and Cost

Electricity is supplied by the local provider Spanish Electricity Company "SWAP ENERGIA". There are six different electricity tariffs, which vary based on the day, the month, the hour, weekends, and holidays. The full tariff scheme of FSM can be found in the picture below.

Figure 5: FSM Tariff Scheme

Hora	Enero	Febrero	Marzo	Abril	Mayo	Junio	Julio	Agosto	Septiembre	Octubre	Noviembre	Diciembre	Sábados, domingos y festivos
0:00 - 1:00	P6	P6	P6	P6	P6	P6	P6	P6	P6	P6	P6	P6	P6
1:00 - 2:00	P6	P6	P6	P6	P6	P6	P6	P6	P6	P6	P6	P6	P6
2:00 - 3:00	P6	P6	P6	P6	P6	P6	P6	P6	P6	P6	P6	P6	P6
3:00 - 4:00	P6	P6	P6	P6	P6	P6	P6	P6	P6	P6	P6	P6	P6
4:00 - 5:00	P6	P6	P6	P6	P6	P6	P6	P6	P6	P6	P6	P6	P6
5:00 - 6:00	P6	P6	P6	P6	P6	P6	P6	P6	P6	P6	P6	P6	P6
6:00 - 7:00	P6	P6	P6	P6	P6	P6	P6	P6	P6	P6	P6	P6	P6
7:00 - 8:00	P6	P6	P6	P6	P6	P6	P6	P6	P6	P6	P6	P6	P6
8:00 - 9:00	P2	P2	P3	P5	P5	P4	P2	P4	P4	P5	P3	P2	P6
9:00 - 10:00	P1	P1	P2	P4	P4	P3	P1	P3	P3	P4	P2	P1	P6
10:00 - 11:00	P1	P1	P2	P4	P4	P3	P1	P3	P3	P4	P2	P1	P6
11:00 - 12:00	P1	P1	P2	P4	P4	P3	P1	P3	P3	P4	P2	P1	P6
12:00 - 13:00	P1	P1	P2	P4	P4	P3	P1	P3	P3	P4	P2	P1	P6
13:00 - 14:00	P1	P1	P2	P4	P4	P3	P1	P3	P3	P4	P2	P1	P6
14:00 - 15:00	P2	P2	P3	P5	P5	P4	P2	P4	P4	P5	P3	P2	P6
15:00 - 16:00	P2	P2	P3	P5	P5	P4	P2	P4	P4	P5	P3	P2	P6
16:00 - 17:00	P2	P2	P3	P5	P5	P4	P2	P4	P4	P5	P3	P2	P6
17:00 - 18:00	P2	P2	P3	P5	P5	P4	P2	P4	P4	P5	P3	P2	P6
18:00 - 19:00	P1	P1	P2	P4	P4	P3	P1	P3	P3	P4	P2	P1	P6
19:00 - 20:00	P1	P1	P2	P4	P4	P3	P1	P3	P3	P4	P2	P1	P6
20:00 - 21:00	P1	P1	P2	P4	P4	P3	P1	P3	P3	P4	P2	P1	P6
21:00 - 22:00	P1	P1	P2	P4	P4	P3	P1	P3	P3	P4	P2	P1	P6
22:00 - 23:00	P2	P2	P3	P5	P5	P4	P2	P4	P4	P5	P3	P2	P6
23:00 - 00:00	P2	P2	P3	P5	P5	P4	P2	P4	P4	P5	P3	P2	P6

Electricity monthly consumptions and costs during the baseline period are highlighted in the table below. The annual electricity consumption of the hotel is 7,148,676 KWh.

Table 2: FSM Monthly Electricity Consumption and Cost

Month	Consumption		Rate	Energy Value	Deviation
Unit	KWh	€	€/KWh	MJ	-
Jan	609,665	92,212 €	0.1513 €	2,194,794	2%
Feb	541,480	73,192 €	0.1352 €	1,949,328	-10%
Mar	627,794	75,674 €	0.1205 €	2,260,058	5%
Apr	563,606	69,797 €	0.1238 €	2,028,982	-6%
May	594,946	85,410 €	0.1436 €	2,141,806	0%
Jun	575,594	95,514 €	0.1659 €	2,072,138	-3%
Jul	632,304	119,253 €	0.1886 €	2,276,294	6%
Aug	646,967	123,875 €	0.1915 €	2,329,081	8%

Month	Consumption		Rate	Energy Value	Deviation
Unit	KWh	€	€/KWh	MJ	-
Sep	611,168	142,347 €	0.2329 €	2,200,205	3%
Oct	618,439	172,044 €	0.2782 €	2,226,380	4%
Nov	567,058	153,191 €	0.2702 €	2,041,409	-5%
Dec	559,655	153,905 €	0.2750 €	2,014,758	-6%
Total	7,148,676	1,356,414 €	0.1897 €	25,735,234	-
Maximum	646,967 (Aug)	172,044 € (Oct)	0.28 € (Oct)	2,329,081 (Aug)	8% (Aug)
Average	595,723	113,034 €	0.1897 €	2,144,603	-
Minimum	541,480 (Feb)	69,797 € (Apr)	0.12 € (Mar)	1,949,328 (Feb)	-10% (Feb)

As it can be seen, average electricity tariff has been increasing since April, which has led to a significant increase in the total electricity cost.

The below figures show the electricity monthly consumption along with the occupancy and CDD respectively during the baseline year.

Figure 6: FSM Electricity Monthly Variations along with the occupancy

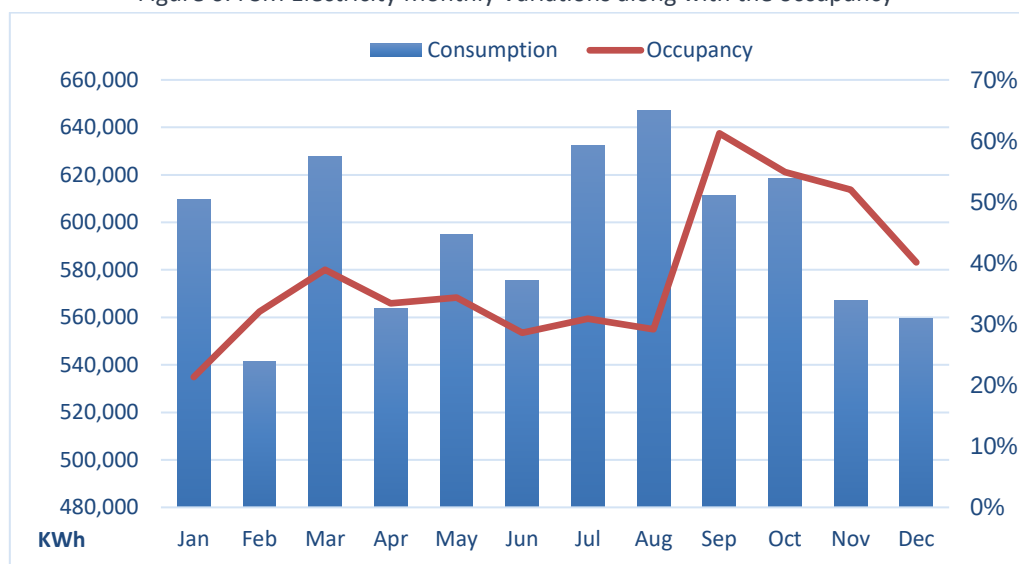
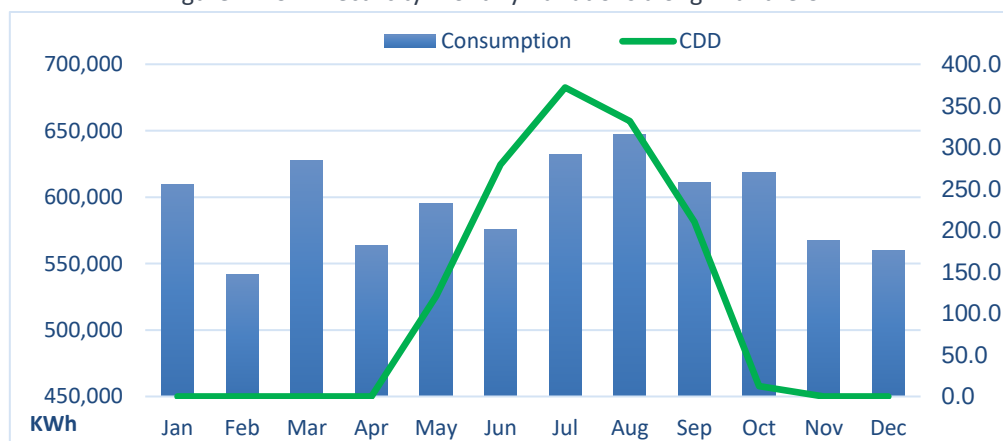


Figure 7: FSM Electricity Monthly Variations along with the CDD



3.1.2. Regression Analysis

In statistical modeling, regression analysis is a set of statistical processes for estimating the relationship among variables. It focuses on the relationship between a dependent variable and one or more independent variables.

In the case of FSM, for a given month, a relationship exists between the energy consumption (the main parameter) and the following variables: the occupancy expressed in Room Nights and the weather conditions (mathematically represented by the cooling degree days). Using the regression analysis, an empirical equation is modeled to generate projected energy consumption as a function of the above-mentioned variables. The model is linear, and the equation will be of the form:

$$\text{KWh} = \text{AxOCC} + \text{BxCDD} + \text{C}$$

OCC is the occupancy rate expressed in Room Nights and CDD is the Cooling Degree Day

A, B and C are the coefficients to be calculated in the process of the regression analysis

In the present regression analysis, the historical figures used for the modeling are found in the below table.

Table 3: Regression Analysis Variables

Month	Occupancy	CDD
Unit	Room Nights	°C.Days
Jan	1,323	0
Feb	1,796	0
Mar	2,414	0
Apr	2,004	0
May	2,130	121
Jun	1,714	279
Jul	1,914	372
Aug	1,807	332
Sep	3,675	210
Oct	3,405	12
Nov	3,120	0
Dec	2,487	0
Total	27,789	1,326

Using these variables, and the energy consumption for the baseline period of each month, the coefficients of the regression model are obtained, leading to the below:

$$\text{KWh(per month)} = (7.1)\text{xOCC} + (125.3)\text{xCDD} + (565,442.1)$$

The projected consumption would be then calculated based on the above equation and the following table provides both the real baseline electricity consumption as well as the projected one if the regression equation is used.

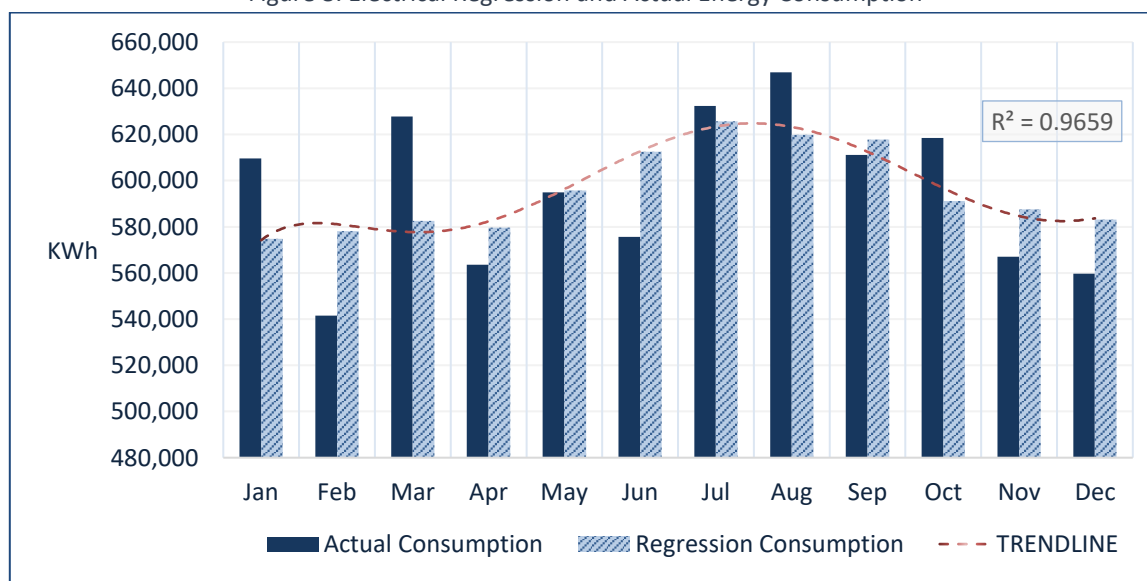
Table 4: Electricity Projected Consumption, Residuals, and variation

Month	Actual Consumption	Regression Consumption	Residuals	Deviation
Unit	KWh	KWh	KWh	%
Jan	609,665	574,829	34,836	6%
Feb	541,480	578,184	-36,704	-7%
Mar	627,794	582,569	45,225	7%
Apr	563,606	579,660	-16,054	-3%
May	594,946	595,709	-763	0%
Jun	575,594	612,575	-36,981	-6%
Jul	632,304	625,652	6,652	1%
Aug	646,967	619,841	27,126	4%
Sep	611,168	617,839	-6,671	-1%
Oct	618,439	591,154	27,285	4%
Nov	567,058	587,578	-20,520	-4%
Dec	559,655	583,087	-23,432	-4%
Total	7,148,676	7,148,676	0	0%

The residuals are the difference between the actual and the projected energy consumption for the same values of independent variables. The variation is the percentage of difference between the actual and the projected consumption. As can be seen, the annual regression provides an exact match (difference of 0); the following chart gives a clear visibility on the regression model and the actual consumption for the baseline period.

R-squared is a statistical measure linked to the regression model. It is also known as the coefficient of determination, which is the percentage of the response variable variation that is explained by a linear model (explained variation / total variation). R-squared is always between 0 and 100%: the higher the value, the better the model fits the data. In the case of FSB, R^2 was seen to be 0.9659, which means that the model can be considered as an acceptable model.

Figure 8: Electrical Regression and Actual Energy Consumption



3.2. Natural Gas

Natural Gas is used in FSM to run some kitchen equipment in the hotel, as well as running the three Viessmann Vitocrossal gas boilers used for domestic hot water production, space heating and pool heating.

Figure 9: Gas Boilers



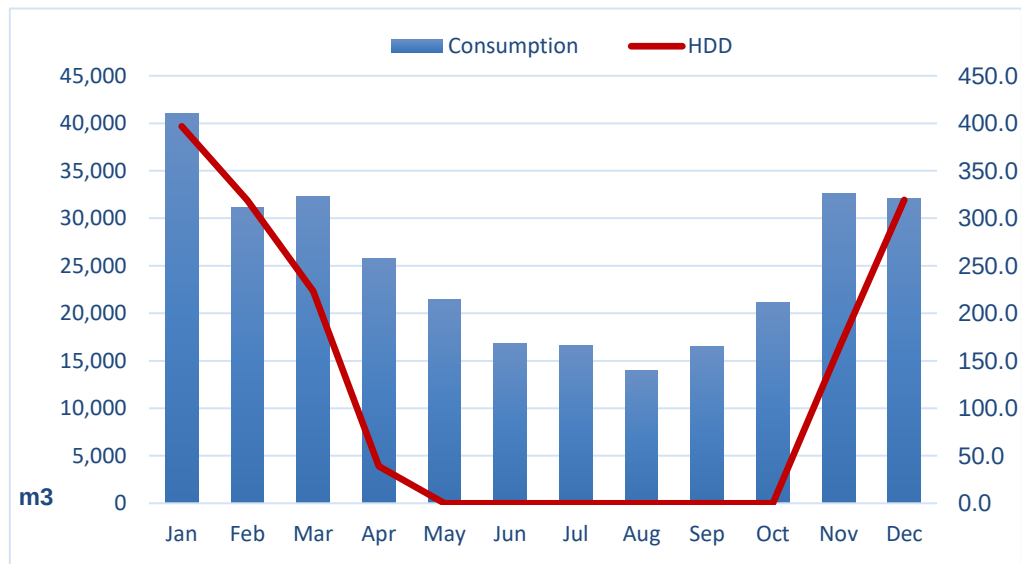
The total audited gas consumption is then **301,520 m³** and accounts for **238,866 €** annually at an average rate of **0.7922 €/m³**. The following table depicts the monthly Gas consumption over the baseline year.

Table 5: FSM Monthly Gas Consumption and Cost

Month	Consumption		Rate	Energy Value	Deviation
Unit	m3	€	€/m3	MJ	-
Jan	41,044	32,569 €	0.7935 €	1,257,588	39%
Feb	31,127	24,700 €	0.7935 €	953,731	19%
Mar	32,334	25,658 €	0.7935 €	990,714	22%
Apr	25,786	20,462 €	0.7935 €	790,083	3%
May	21,456	17,026 €	0.7935 €	657,412	-17%
Jun	16,874	13,390 €	0.7935 €	517,019	-49%
Jul	16,614	13,184 €	0.7935 €	509,053	-51%
Aug	13,934	11,057 €	0.7935 €	426,938	-80%
Sep	16,490	13,085 €	0.7935 €	505,254	-52%
Oct	21,104	16,746 €	0.7935 €	646,627	-19%
Nov	32,626	25,889 €	0.7935 €	999,661	23%
Dec	32,131	25,101 €	0.7812 €	984,494	22%
Total	301,520	238,866 €	0.7922 €	9,238,573	-
Maximum	41,044 (Jan)	32,569 € (Jan)	0.7935 € (Jan)	1,257,588 (Jan)	39% (Jan)
Average	25,127	19,906 €	0.7925 €	769,881	-
Minimum	13,934 (Aug)	11,057 € (Aug)	0.7812 € (Dec)	426,938 (Aug)	-80% (Aug)

As it can be seen in the graph below, the gas consumption is directly correlated to the HDD (Heating Degree Days) since the gas consumptions increases during winter months where the HDD values are high compared to summertime.

Figure 10: FSM Gas Monthly Variations vs. HDD



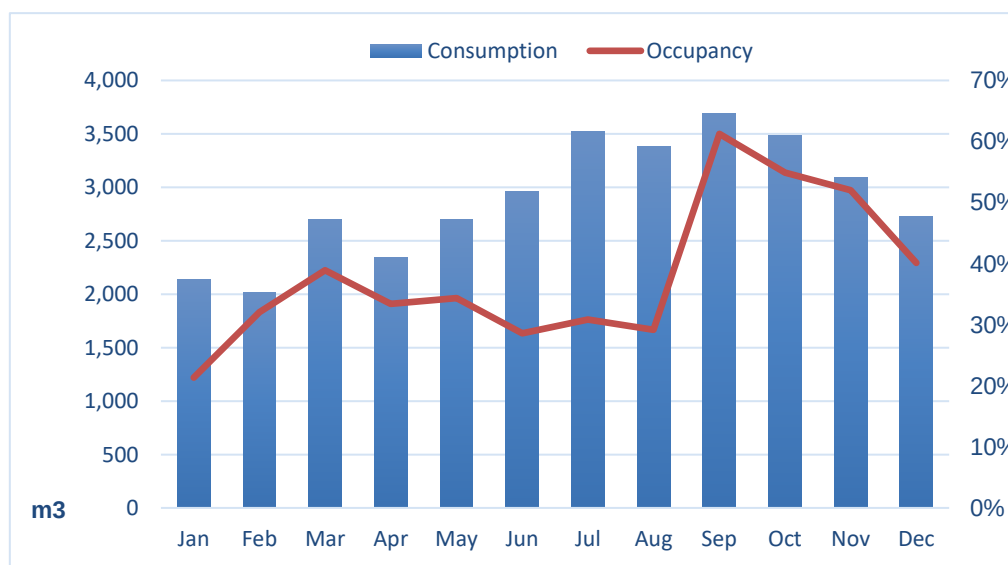
3.3. Water

Water is supplied to FSM by the National Water Company for domestic use. The total water consumption for the baseline year is shown in the following table and chart.

Table 6: FSM Monthly Water Consumption and Cost

Month	Consumption		Rate	Deviation
Unit	m3	€	€/m3	-
Jan	2,138	4,148 €	1.94 €	-35%
Feb	2,013	4,026 €	2.00 €	-44%
Mar	2,699	5,398 €	2.00 €	-7%
Apr	2,341	4,916 €	2.10 €	-24%
May	2,702	5,674 €	2.10 €	-7%
Jun	2,963	7,141 €	2.41 €	2%
Jul	3,522	8,488 €	2.41 €	18%
Aug	3,386	6,806 €	2.01 €	14%
Sep	3,689	7,415 €	2.01 €	21%
Oct	3,486	7,007 €	2.01 €	17%
Nov	3,093	6,217 €	2.01 €	6%
Dec	2,726	5,725 €	2.10 €	-6%
Total	34,758	72,960 €	2.10 €	-
Maximum	3,689 (Sep)	8,488 € (Jul)	2.41 € (Jun)	21% (Sep)
Average	2,897	6,080 €	2.09 €	-
Minimum	2,013 (Feb)	4,026 € (Feb)	1.94 € (Jan)	-44% (Feb)

Figure 11: FSM Water Consumption Monthly Variations



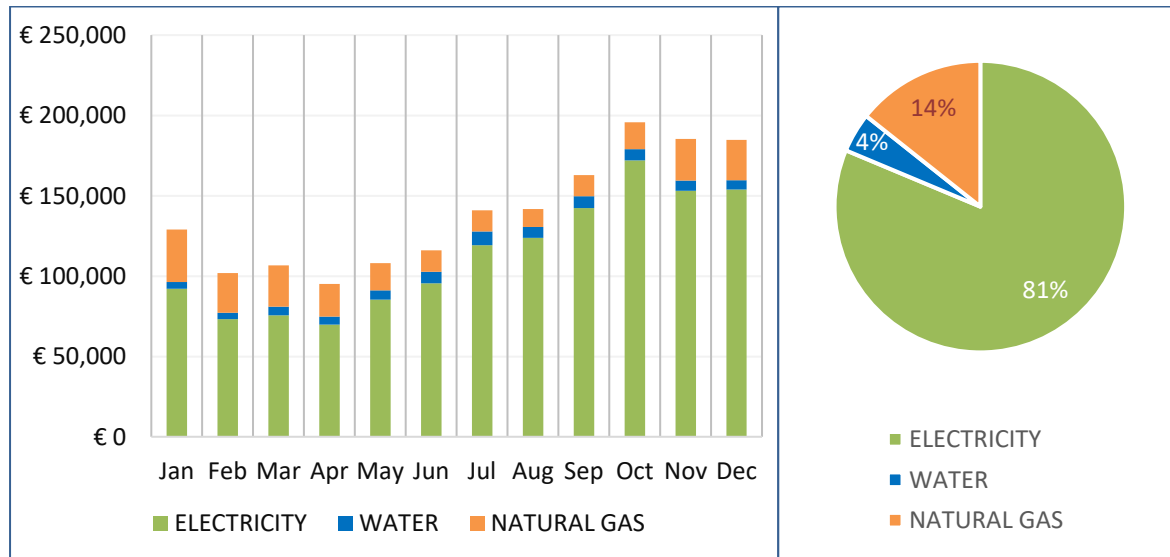
3.4. Utilities Cost Breakdown

The below table provides the monthly cost of each utility supply during for the baseline year of 2021.

Table 7: Utilities Cost Breakdown

MONTH	ELECTRICITY	WATER	NATURAL GAS	TOTAL
Unit	€	€	€	€
Jan	€ 92,212	€ 4,148	€ 32,569	€ 128,929
Feb	€ 73,192	€ 4,026	€ 24,700	€ 101,918
Mar	€ 75,674	€ 5,398	€ 25,658	€ 106,730
Apr	€ 69,797	€ 4,916	€ 20,462	€ 95,175
May	€ 85,410	€ 5,674	€ 17,026	€ 108,110
Jun	€ 95,514	€ 7,141	€ 13,390	€ 116,045
Jul	€ 119,253	€ 8,488	€ 13,184	€ 140,924
Aug	€ 123,875	€ 6,806	€ 11,057	€ 141,738
Sep	€ 142,347	€ 7,415	€ 13,085	€ 162,847
Oct	€ 172,044	€ 7,007	€ 16,746	€ 195,797
Nov	€ 153,191	€ 6,217	€ 25,889	€ 185,297
Dec	€ 153,905	€ 5,725	€ 25,101	€ 184,730
Total	€ 1,356,414	€ 72,960	€ 238,866	€ 1,668,240
Maximum	€172,044 (Oct)	€8,488 (Jul)	€32,569 (Jan)	€195,797 (Oct)
Average	€ 113,034	€ 6,080	€ 19,906	€ 139,020
Minimum	€69,797 (Apr)	€4,026 (Feb)	€11,057 (Aug)	€95,175 (Apr)
Percentage	81%	4%	14%	100%

Figure 12: Utilities Cost Breakdown



As it can be seen, electricity represents the biggest share of the hotel's total utility bill with 81%, followed by Natural Gas at 14% and finally Water at only 4%.

3.5. Energy and Carbon Footprint

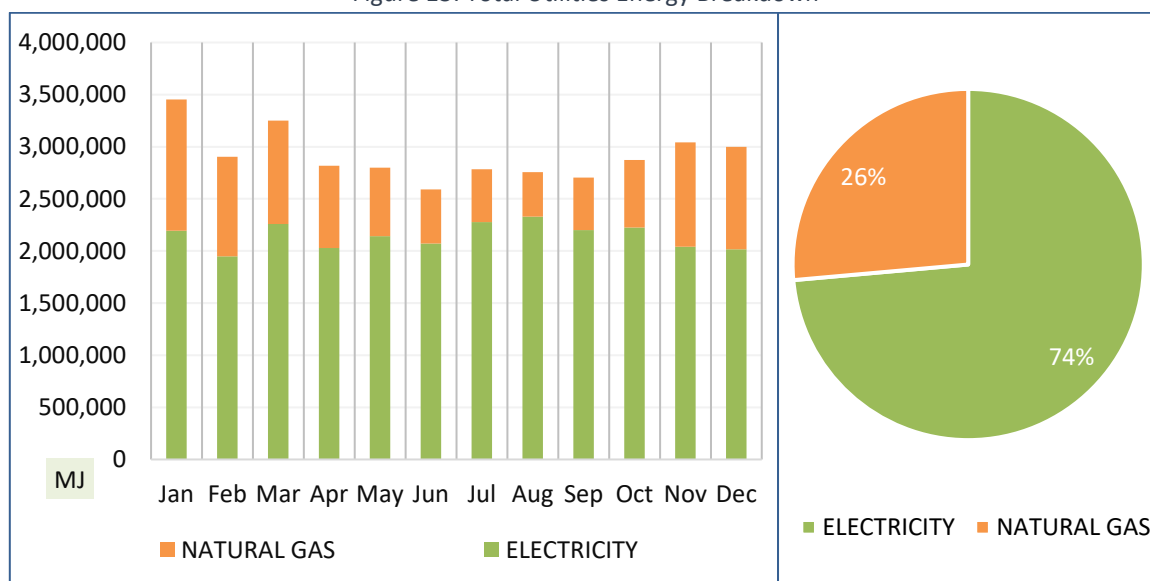
3.5.1. Energy Footprint

To be able to aggregate all the utilities energy consumption, the energy consumption of all energy sources is converted to a single energy unit MJ (mega joules). The breakdown for the energy sources consumption is shown in the following table. As can be seen, electricity accounts for 74% of the energy sources energy consumption, followed by natural gas at 30%, electricity at 26%.

Table 8: Total Utilities Energy Breakdown

MONTH	ELECTRICITY	NATURAL GAS	TOTAL
Unit	MJ	MJ	MJ
Jan	2,194,794	1,257,588	3,452,382
Feb	1,949,328	953,731	2,903,059
Mar	2,260,058	990,714	3,250,772
Apr	2,028,982	790,083	2,819,065
May	2,141,806	657,412	2,799,217
Jun	2,072,138	517,019	2,589,158
Jul	2,276,294	509,053	2,785,347
Aug	2,329,081	426,938	2,756,019
Sep	2,200,205	505,254	2,705,458
Oct	2,226,380	646,627	2,873,007
Nov	2,041,409	999,661	3,041,069
Dec	2,014,758	984,494	2,999,252
Total	25,735,234	9,238,573	34,973,806
Maximum	2,329,081 (Aug)	1,257,588 (Jan)	3,452,382 (Jan)
Average	2,144,603	769,881	2,914,484
Minimum	1,949,328 (Feb)	426,938 (Aug)	2,589,158 (Jun)
Percentage	74%	26%	100%

Figure 13: Total Utilities Energy Breakdown



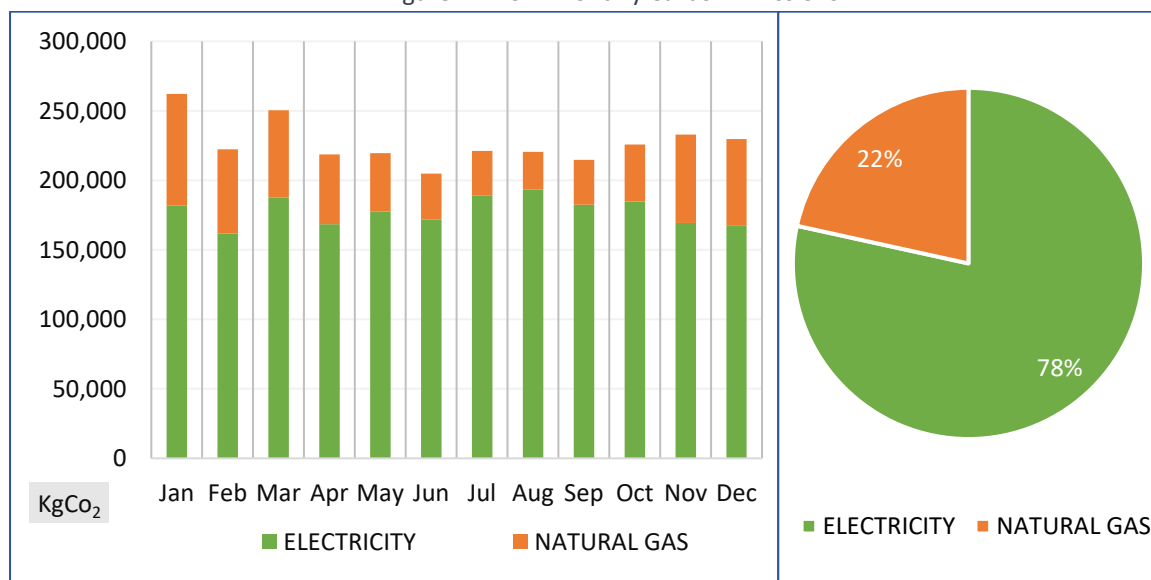
3.5.2. Carbon Emissions

The carbon emissions calculation is done by multiplying the consumption of each utility (electricity, district cooling, district heating and gas) by the emissions factor of each. The electricity emissions factor varies between countries, with the Spain average of 0.2988 KgCO₂/KWh being taken in this report. Natural Gas emission factor is 1.9474 KgCO₂/m³. The **total annual carbon emissions** are at a total of **2,723,090 KgCO₂**.

Table 9: FSM Carbon Emissions Monthly Breakdown

MONTH	ELECTRICITY	NATURAL GAS	TOTAL
Unit	KgCO ₂	KgCO ₂	KgCO ₂
Jan	182,158	79,929	262,087
Feb	161,786	60,617	222,402
Mar	187,575	62,967	250,542
Apr	168,396	50,216	218,612
May	177,760	41,783	219,544
Jun	171,978	32,860	204,839
Jul	188,922	32,354	221,276
Aug	193,303	27,135	220,438
Sep	182,607	32,113	214,720
Oct	184,780	41,098	225,878
Nov	169,428	63,536	232,964
Dec	167,216	62,572	229,788
Total	2,135,910	587,180	2,723,090
Maximum	193,303 (Aug)	79,929 (Jan)	262,087 (Jan)
Average	177,993	48,932	226,924
Minimum	161,786 (Feb)	27,135 (Aug)	204,839 (Jun)
Percentage	78%	22%	100%

Figure 14: FSM Monthly Carbon Emissions



4. Electrical Energy Load Analysis & Energy Balance

4.1. Electrical Load Categories

A crucial milestone in energy audits is the buildup of the energy balance. This process starts with acquiring the load inventory – electrical end uses, and then running a complete simulation based on the operational schedule, load factors and behavioral statistics of each item in the inventory. The micro and macro energy consumption analysis of energy systems including lighting, heating, air conditioning, ventilation and building envelope, serves the purpose of identifying where and how a facility uses energy. The breakdown of the energy consumption by end-use would be the energy balance, which will provide an important baseline for the roadmap in developing the Energy Efficiency Measures (EEMs) that would optimize the overall energy costs.

Information and Database Process

FSM's Load inventory data is populated into an Energy Analysis System including assets name/category, actual electrical specification (KW) and operational schedule over 12 months (hours per day, week, month, or season). The second step is performing a total simulation and energy modeling of the facility with full bills reconciliation as per the baseline values.

In the present report, electrical loads were divided into the following categories:

1- Lighting

This category corresponds to all types of existing lighting fixtures.

2- Cooling

This category corresponds to all chillers and split units in the facility.

3- HVAC Fans

This category corresponds to all extraction fans, fan coil units and air handling units responsible for the required circulation of air in the facility.

4- Motors

This category corresponds to all chilled water pumps (primary and secondary), water pumps (hot and domestic), circulation, pool pumps, and lifts.

5- Kitchen and Refrigeration

This category corresponds to all electricity driven Kitchen and Refrigeration equipment.

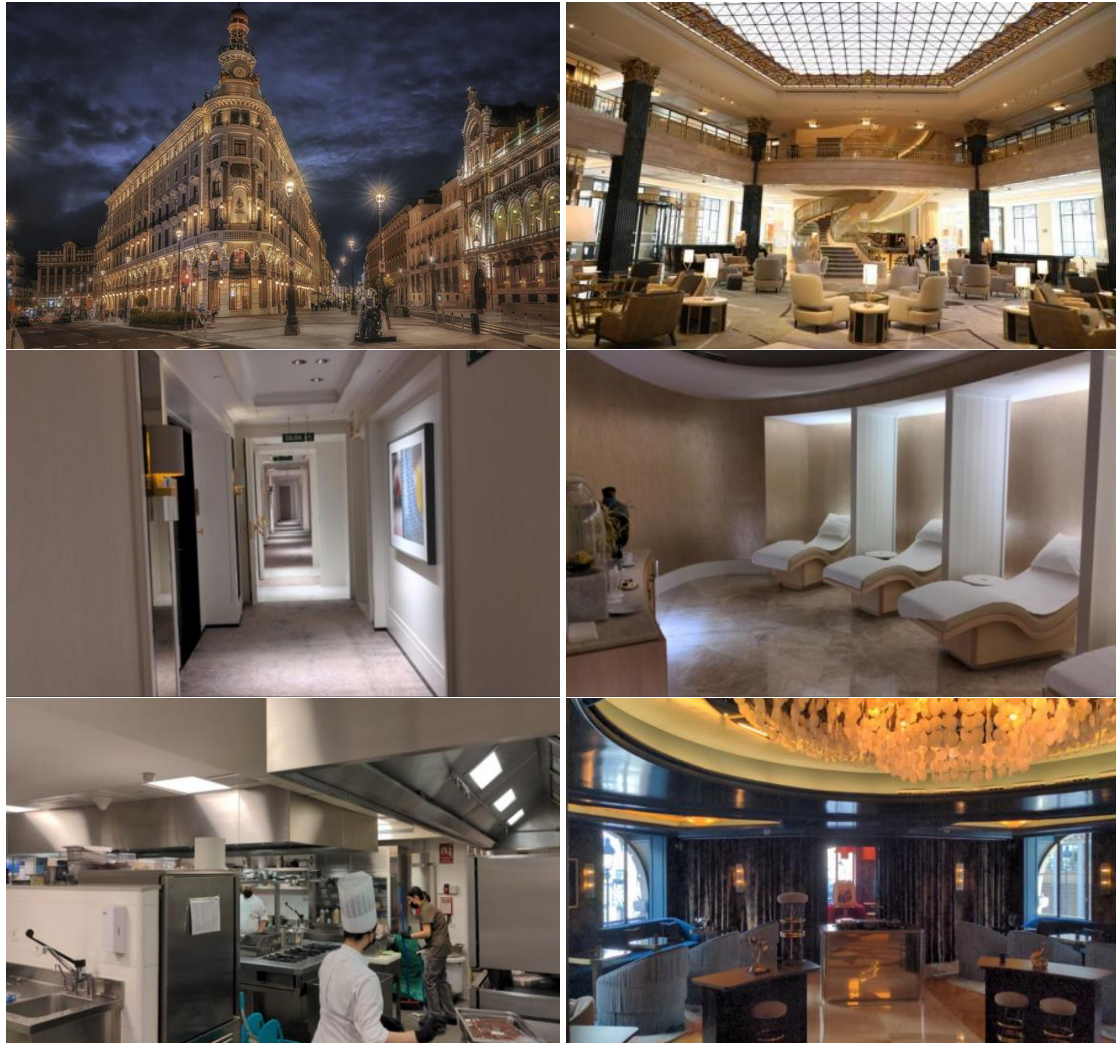
6- Others

This category corresponds to all electrical driven equipment such as IT loads, CCTV, and other miscellaneous equipment.

4.1.1. Lighting

This category corresponds to all types of existing lighting fixtures, which in the case of FSM consists fully of LED fixtures. Since the hotel is hooting to be LEED certified, it is mandatory to install LED fixtures in all the premises of the property. Below are pictures of a few examples of the lighting in FSM.

Figure 15: FSM Lighting



- Top: Façade Lighting (Left); Lobby Entrance (Right)
- Middle: Guest Room Corridor (Left); SPA Treatment Room (Right)
- Bottom: Kitchen (Left); Isa Lounge (Right)

The total **Lighting** category electrical consumption for the baseline period is **523,651 KWh**.

4.1.2. Cooling

This energy end use category corresponds to the three chillers, three cooling towers, the two heat pumps and the few VRVs responsible for space cooling in FSM.

The largest part of the cooling demand is met by the three CARRIER water-cooled chillers along with three cooling towers, one of each chiller, installed on the roof of the hotel. As it can be seen in the picture below, there are six cooling towers installed on the roof, three of which belong to the hotel, with the remaining three serving the department store which is in the same real estate but not part of the hotel, therefore not in the scope of this report.

Figure 16: Chillers (Left) and Cooling Towers (Right)



During winter, when the cooling demand is at its lowest, the chillers and cooling towers system is completely shut down, and chilled water is provided by two AQUASNAP heat pumps to reduce the electricity consumption of the cooling system. The two heat pumps are also equipped with a heat recovery system used for domestic hot water production.

Figure 17: Heat Pumps



As mentioned earlier, there are also a few VRV systems installed on the roof of the hotel serving specific locations such as the IT rooms which are served by 3 VRVs and the communication rooms which are served by 5 VRVs.

Figure 18: IT Rooms (Left) and Communication Rooms (Right) VRVs



The total **Cooling** category electrical consumption for the baseline period is **1,318,933 KWh**.

4.1.3. HVAC Fans

This category includes all extraction fans, kitchen hoods, fan coil units and air handling units responsible for the proper Indoor Air Quality of the facility.

A total of 25 Air Handling Units are installed in FSM. These units vary in size based on the area and nature of the conditioned space. 18 of the 25 AHUs are also equipped with humidifiers to increase the relative humidity of the conditioned spaces, since the property has long suffered from a humidity problem. Many units were seen to have very big humidifiers, with an installed load of approximately 37KW, and what is surprising is that during the field audit, most of the units were operating at 100% load while the outside RH was around 50%, which is well within the comfort level.

Figure 19: AHUs (Left) and Humidifier (Right)



FCUs are also installed in most of the property's spaces while the key extractor fans are the kitchen hoods, toilets extract fans and the rest being small ventilation fans for proper air circulation and rejection.

Figure 20: Kitchen Hoods Exhaust



The total **HVAC Fans** category electrical consumption for the baseline period is **2,485,133 KWh**.

4.1.4. Motors

This end use category includes all pumps – primary chilled water, secondary chilled water, booster, heat recovery, return, pool, circulation, and hot water pumps, as well as all the service lifts and guest elevators.

Figure 21: FSM Pumps



The total **Motors** category electrical consumption for the baseline period is **1,695,196 KWh**.

4.1.5. Kitchen and Refrigeration

This category includes all electricity driven Kitchen and Refrigeration equipment across all the F&B facilities and kitchens in the premises.

Figure 22: Kitchen Equipment (Left) and Cold Room Compressors (Right)



The total **Kitchen and Refrigeration** category electrical consumption for the baseline period is **591,732 KWh**.

4.1.6. Others

This category corresponds to all the remaining electrical driven equipment such as IT loads, CCTV, Gym equipment and other miscellaneous equipment.

Figure 23: Gym Equipment (Left) and IT Loads (Right)



The total **Others** category electrical consumption for the baseline period is **542,197 KWh**.

4.2. Energy Balance

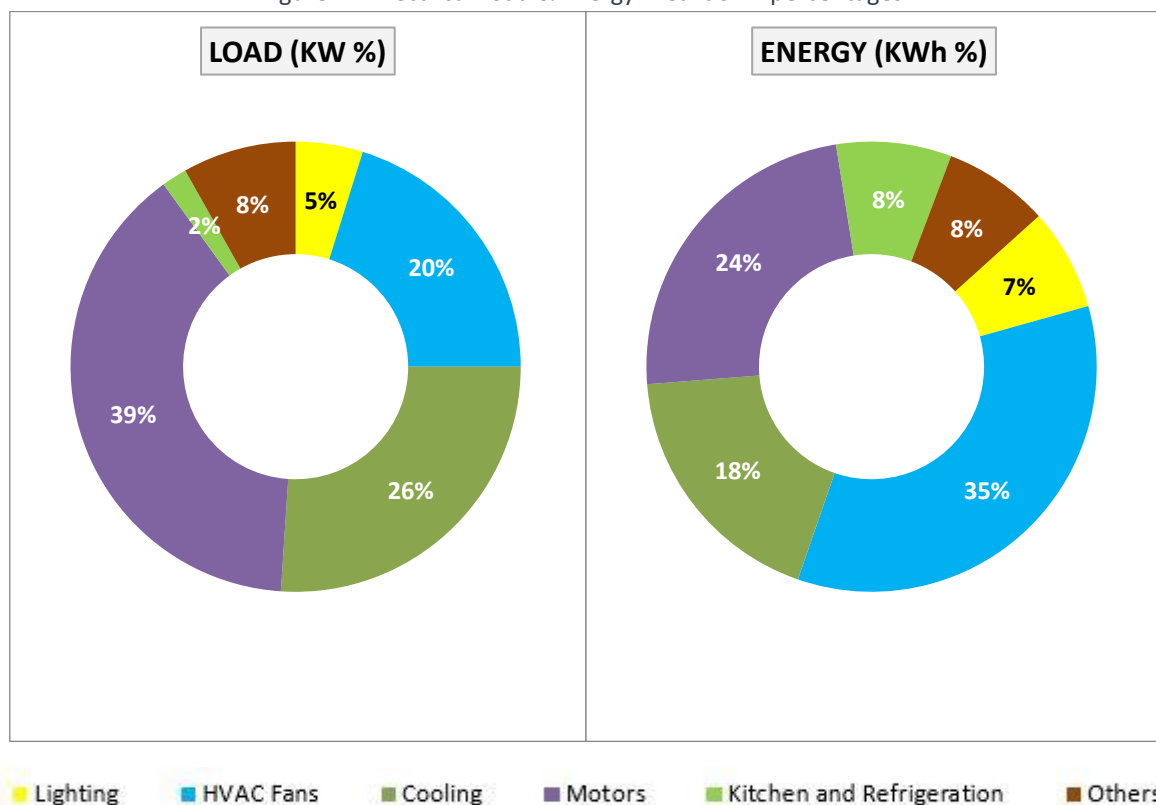
The Electrical Energy Balance provides the breakdown per chosen end-use category for both the installed electrical loads (KW) and the annual energy consumed (KWh). This is achieved following the energy simulation and bills reconciliation completion as per the load inventory, the related operations and the inspection done during the field energy audit.

The following table and figure provide the percentage shares of categories' total installed electrical power in KW and energy consumption in KWh.

Table 10: Electrical Load & Energy Consumption Breakdown by Category

Category	Load		Energy			
Unit	KW	%	KWh	€	Kg. CO ₂	%
Lighting	133	5%	523,651	€ 99,337	156,467	7%
HVAC Fans	556	20%	2,476,967	€ 469,881	740,118	35%
Cooling	718	26%	1,318,933	€ 250,202	394,097	18%
Motors	1,074	39%	1,695,196	€ 321,579	506,524	24%
Kitchen and Refrigeration	50	2%	591,732	€ 112,252	176,810	8%
Others	225	8%	542,197	€ 102,855	162,009	8%
Total	2,757	100%	7,148,676	€ 1,356,104	2,136,024	100%
Maximum	Motors - 1,074, 39%		HVAC Fans - 2,476,967 KWh, 469,881, 35%			
Minimum	Kitchen and Refrigeration - 50, 2%		Lighting - 523,651 KWh, 99,337, 7%			

Figure 24: Electrical Load & Energy Breakdown percentages.



5. Energy Efficiency Measures

The main objective of this section is to provide a roadmap that would include several Energy Efficiency Measures (EEM) which are all the actions that would reduce the utilities consumption. In this aim, EEG investigated a considerable number of energy efficiency solutions, which could yield to substantial energy cost and carbon footprint reductions in the event of their implementation.

These EEMs are all in the category of the 'LCNC – Low Cost/No Cost' type and can be implemented seamlessly by the engineering department. As can be seen, annual cost savings in excess of €275,000 are possible.

The list of EEMs that are developed in this section are found in the below table along with the note that the tariff negotiation/demand review was not considered in the cost savings as well as the lighting management EEM.

Table 11: EEMs for FSM with KWh , Cost and KgCO₂ savings

CODE	EEM Name	ELECTRICITY	ANNUAL SAVINGS	
		KWh	€	KGs CO ₂
-	-			
EEM - 01 [LCNC]	Review subscribed electricity demand levels	0 (0.0%)	0 € (0.0%)	0
EEM - 02 [LCNC]	Lighting Management in Function Areas	0 (0.0%)	0 € (0.0%)	0
EEM - 03 [LCNC]	HVAC Control: Aggressive scheduling	161,271 (2.3%)	30,600 € (1.8%)	48,185
EEM - 04 [LCNC]	HVAC Control: Humidifiers' operation	999,519 (14.0%)	189,652 € (11.4%)	298,640
EEM - 05 [LCNC]	HVAC Control: Review all pressure setpoints Control of VFDs	215,028 (3.0%)	40,800 € (2.4%)	64,247
EEM - 06 [LCNC]	Chillers/Heat Pumps: Chilled Water Temperature Reset	27,968 (0.4%)	5,307 € (0.3%)	8,356
EEM - 07 [LCNC]	Chilled Water: Proactive ΔP Setpoint reset at Secondary Chilled Water Pumps	49,952 (0.7%)	9,478 € (0.6%)	14,925
	TOTAL	1,453,739 20%	275,837 € (16.5%)	434,354

5.1. EEM-01: Review subscribed electricity demand levels

Currently, FSM has a contract with the electricity utility SWAP Energia that is based on various cost structures, notably divided in 6 different Time Of Use (TOU) tariffs and subscribed power. The breakdown of usage of each TOU can be seen in the below weekly schedule.

Figure 25: the 6 Time of Use Slots of the electric utility

Hora	Enero	Febrero	Marzo	Abril	Mayo	Junio	Julio	Agosto	Septiembre	Octubre	Noviembre	Diciembre	Sábados, domingos y festivos
0:00 - 1:00	P6	P6	P6	P6	P6	P6	P6	P6	P6	P6	P6	P6	P6
1:00 - 2:00	P6	P6	P6	P6	P6	P6	P6	P6	P6	P6	P6	P6	P6
2:00 - 3:00	P6	P6	P6	P6	P6	P6	P6	P6	P6	P6	P6	P6	P6
3:00 - 4:00	P6	P6	P6	P6	P6	P6	P6	P6	P6	P6	P6	P6	P6
4:00 - 5:00	P6	P6	P6	P6	P6	P6	P6	P6	P6	P6	P6	P6	P6
5:00 - 6:00	P6	P6	P6	P6	P6	P6	P6	P6	P6	P6	P6	P6	P6
6:00 - 7:00	P6	P6	P6	P6	P6	P6	P6	P6	P6	P6	P6	P6	P6
7:00 - 8:00	P6	P6	P6	P6	P6	P6	P6	P6	P6	P6	P6	P6	P6
8:00 - 9:00	P2	P2	P3	P5	P5	P4	P2	P4	P4	P5	P3	P2	P6
9:00 - 10:00	P1	P1	P2	P4	P4	P3	P1	P3	P3	P4	P2	P1	P6
10:00 - 11:00	P1	P1	P2	P4	P4	P3	P1	P3	P3	P4	P2	P1	P6
11:00 - 12:00	P1	P1	P2	P4	P4	P3	P1	P3	P3	P4	P2	P1	P6
12:00 - 13:00	P1	P1	P2	P4	P4	P3	P1	P3	P3	P4	P2	P1	P6
13:00 - 14:00	P1	P1	P2	P4	P4	P3	P1	P3	P3	P4	P2	P1	P6
14:00 - 15:00	P2	P2	P3	P5	P5	P4	P2	P4	P4	P5	P3	P2	P6
15:00 - 16:00	P2	P2	P3	P5	P5	P4	P2	P4	P4	P5	P3	P2	P6
16:00 - 17:00	P2	P2	P3	P5	P5	P4	P2	P4	P4	P5	P3	P2	P6
17:00 - 18:00	P2	P2	P3	P5	P5	P4	P2	P4	P4	P5	P3	P2	P6
18:00 - 19:00	P1	P1	P2	P4	P4	P3	P1	P3	P3	P4	P2	P1	P6
19:00 - 20:00	P1	P1	P2	P4	P4	P3	P1	P3	P3	P4	P2	P1	P6
20:00 - 21:00	P1	P1	P2	P4	P4	P3	P1	P3	P3	P4	P2	P1	P6
21:00 - 22:00	P1	P1	P2	P4	P4	P3	P1	P3	P3	P4	P2	P1	P6
22:00 - 23:00	P2	P2	P3	P5	P5	P4	P2	P4	P4	P5	P3	P2	P6
23:00 - 00:00	P2	P2	P3	P5	P5	P4	P2	P4	P4	P5	P3	P2	P6

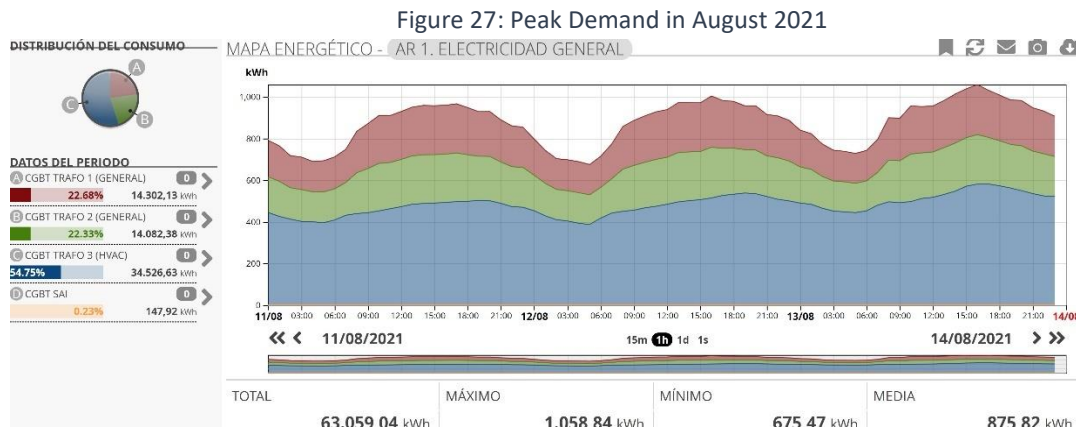
The billing structure includes as mentioned a subscribed power fee for each of these TOU slot that is charged according to the number of days per month (daily charges x # Days of the month). Below is an example of a 30 days month whereby the total charges top the €13,200

Figure 26: TOU slots charges in November 2021 bill (30 days)

P1 1600,000 kW x 30 Días x 0,086238 €/kW día	4.139,42 €
P2 1600,000 kW x 30 Días x 0,073523 €/kW día	3.529,10 €
P3 1600,000 kW x 30 Días x 0,043385 €/kW día	2.082,48 €
P4 1600,000 kW x 30 Días x 0,035674 €/kW día	1.712,35 €
P5 1600,000 kW x 30 Días x 0,013333 €/kW día	639,98 €
P6 3200,000 kW x 30 Días x 0,008320 €/kW día	798,72 €

During the field audit, an analysis of the real power demand (in KW) was made on the existing Energy Management System Platform, and it was clearly seen that based on the present operation, the maximum demand is not exceeding the 1050KW in peak summer months which would be the higher electrical demand time (due to the chillers operation).

The following chart showcases the August peak power demand,



On the other hand, at the present time, all subscribed limits are 1600KW except for P6 TOU which is 3200KW.

Based on the energy analysis and measurements taken from the EMS, it is recommended to reduce the subscribed power in all TOU slots from their present levels to 1200KW as a first stage.

The **total savings to be achieved annually will top the €38,000** as seen in the below table.

Table 12: Sea water condenser cooling project Cooling Towers Water & RO savings

Time of Use	Current Limit (KW)	Tariff (€/KW/Day)	New Limit (KW)	Annual Savings
P1	1600	0.086238	1200	€ 12,591
P2	1600	0.073523	1200	€ 10,734
P3	1600	0.043385	1200	€ 6,334
P4	1600	0.035674	1200	€ 5,208
P5	1600	0.013333	1200	€ 1,947
P6	3200	0.00832	1200	€ 1,215
TOTAL Savings				€ 38,029

It is crucial to note that with the combination of EMS and BMS platforms available at the site, it is very possible to build an automatic load shedding system should the power demand exceeds a threshold over a set time.

For example, if the power demand exceeds the subscribed power of 1200KW for 5 consecutive minutes, a load shedding trigger will shut down one of the chillers for a certain time until the levels drop to another set threshold. This will ensure FSM to comply 100% with any subscribed power demand it chooses.

The 1200KW are however easily achieved as currently the maximum levels topped barely the 1050KW while there are substantial power reductions possible as can be seen in the forthcoming sections.

Apart the subscribed maximum demand (KW), it is to be noted that P6 TOU also carries the lowest energy tariff (€/KWh) with up to 11% lower as compared to the P2 slot. Accordingly, a

Careful analysis of the loads operating during P2 slot should be made and accordingly plan whenever possible either a load shedding or a load rescheduling to P6 slot. One example would be electrical ovens in kitchen that could be rescheduled to a few hours before or later, or else all HVAC loads that can be controlled.

Shifting 10,000 KWh only from P2 to P6 slot would lead to annual savings close to €4,000.

5.2. EEM-02: Lighting Management in Function Areas

The function areas are a total of about 1500m² and include a large lighting load. It was noticed that all lights are kept operating during the preparation of the rooms where events will occur and at the time of dismantling any installations or removing the setup.

It is recommended to coordinate with the F&B department to ensure:

- i. All functions rooms to have their lighting OFF during vacancy
- ii. All function rooms used for an event to have either its lighting partially shut down or better to implement a dimming strategy on all lighting during both the preparation of the room(s) and the removal of all setup.

Figure 28: Largest ballroom during an event's setup removal



5.3. EEM-03: HVAC Control: Aggressive scheduling

All AHUs in FSM are controlled via a schedule of operation set from the BMS. They are mostly fixed across the week. The below two examples show the schedule of operation of all function areas' AHUs such as the AHU Ballroom P2 8B and the ISA Blue Room in the following page.

The AHUs of the ballrooms/functions areas are all following the schedule of 24/7 operation except for Sundays while ISA blue room operated from 7hAM to 1hAM daily regardless of its occupancy.

Figure 29: AHU Ballroom P2 8B Schedule of operation

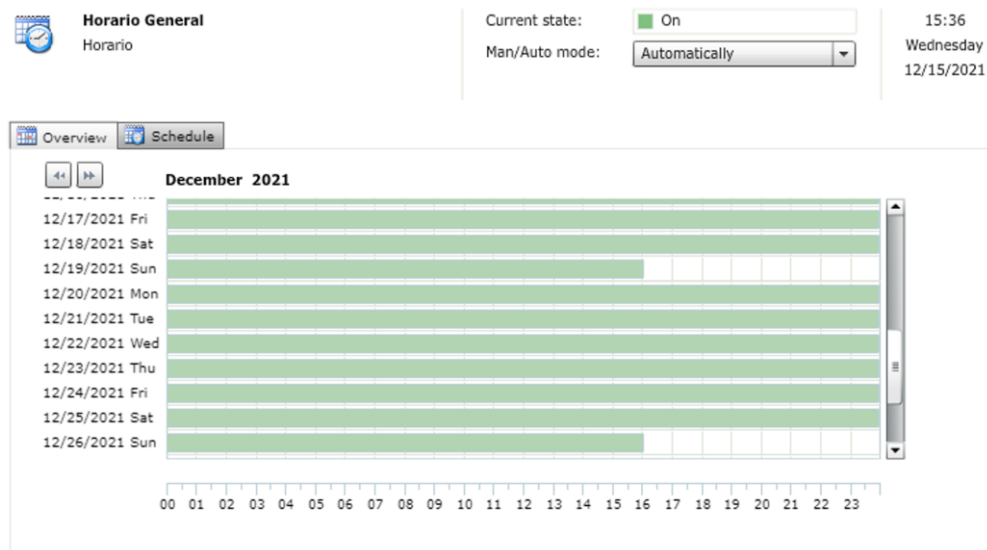
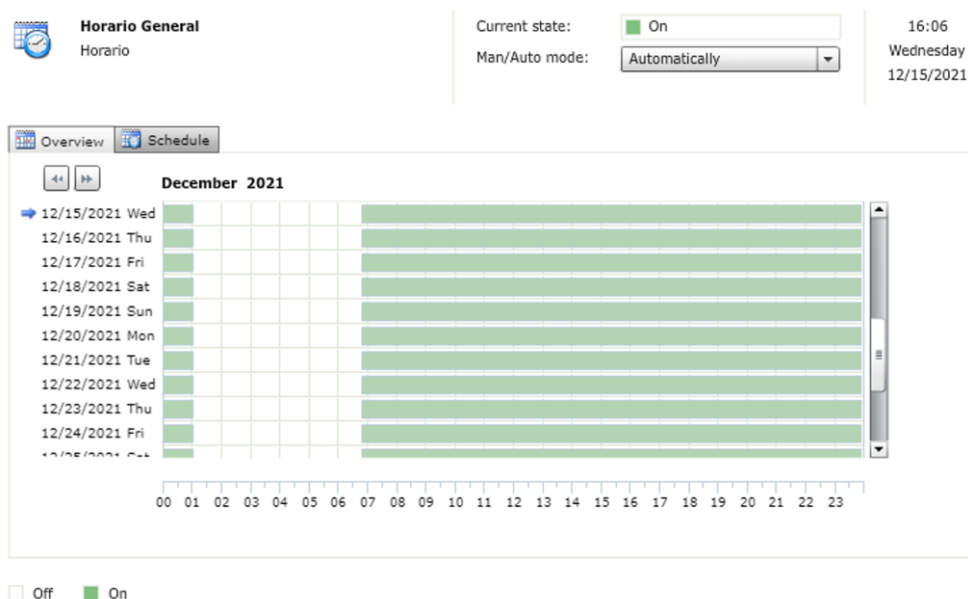


Figure 30: ISA Blue Room AHU schedule of operation



The key mitigation actions are the following:

- Review the operation of each AHU as well as the area it services.
- Implement an aggressive scheduling that should be proactively modified every day in some units, every week in others. This includes the below two scheduling possibilities:
 - I. Some units can be shut down totally nighttime or during vacancy such as all those serving the ballrooms and functions areas. During the day they can be duty cycled ON/OFF if also vacant (i.e: 1 hour ON/ 1 hour OFF) if needed. This would be better than maintaining them operational continuously.
 - II. Some units that can't be shut down fully at nighttime, can be duty cycled in low occupancy period such as nighttime or known low circulation times. Duty cycling can vary from unit to unit and can be as low as 45mn ON/15mn OFF to larger window of time.

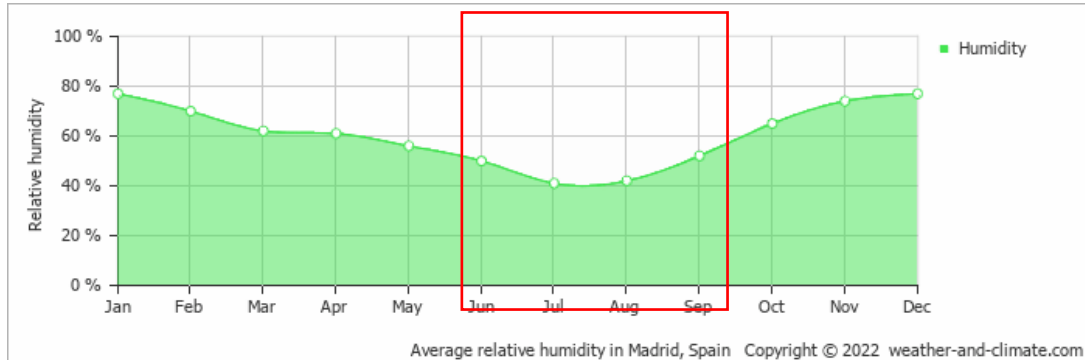
Savings will be a minimum of 15% on the total Fans operation of AHUs. It will be calculated in the final energy audit report but **should easily top the 150,000 KWh annually equivalent to €41,250 at the latest electricity tariff of €0.275/KWh.**

EEM - 03		
HVAC Control: Aggressive scheduling		
ENERGY SAVINGS		
ELECTRICITY	GAS	CO ₂
KWh	m3	Kg
161,271	0	48,185
2.3%	0.0%	1.8%
ECONOMIC PERFORMANCE		
Annual Cost Savings [€]		30,600 €
Annual Share of Cost Savings [%]		1.8%
Implementation Cost [€]		0 €

5.4. EEM-04: HVAC Control: Humidifiers' operation

The average annual relative humidity is 61.8% and average monthly relative humidity ranges from 42% in July to 79% in December with the summer months being the driest months.

Figure 31: ISA Blue Room AHU schedule of operation



The HVAC design of the ventilation system of FSM includes an electrical driven humidifier on most AHUs to ensure a minimum level of humidity is attained.

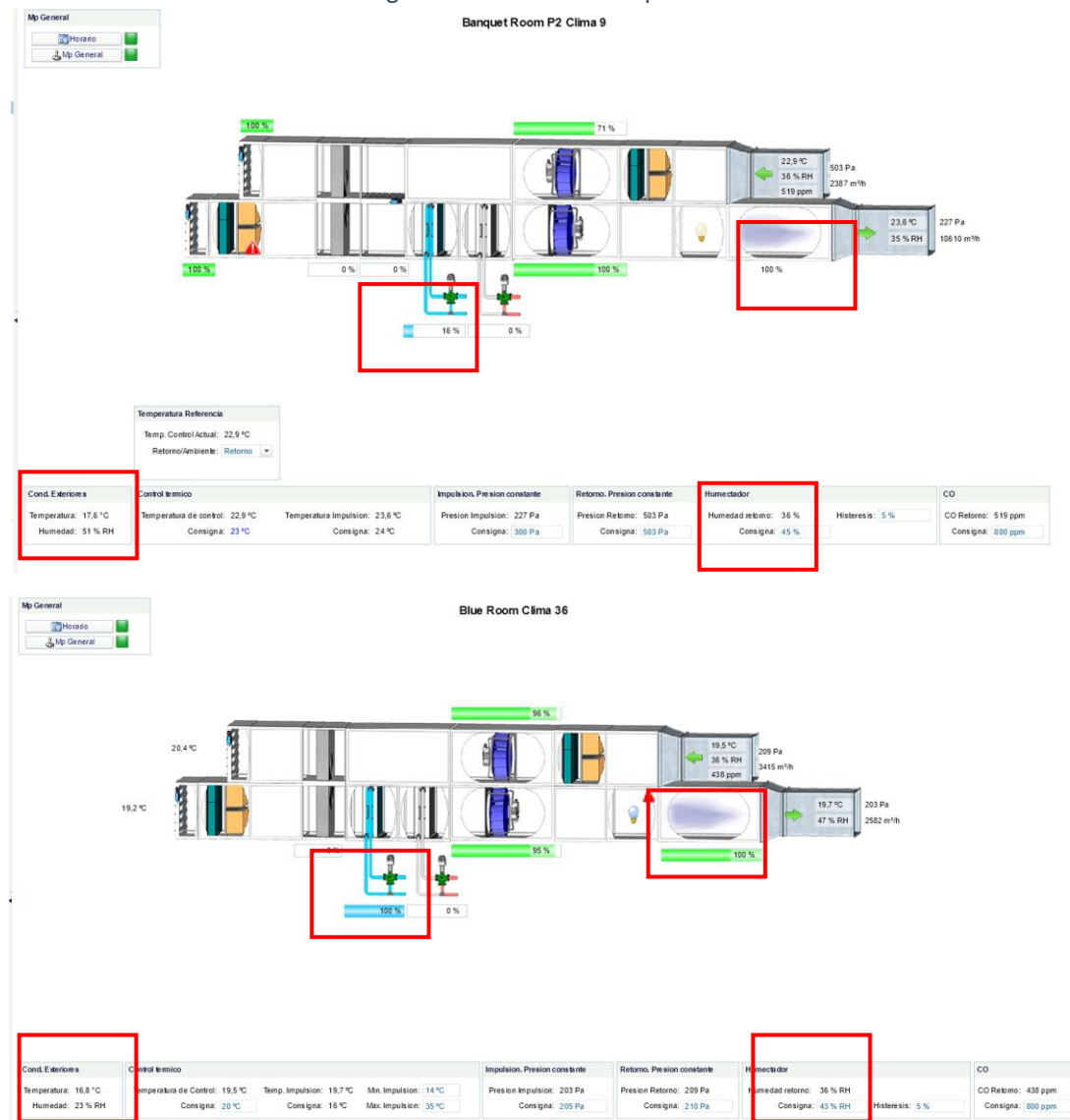
Upon analyzing all the operation of the HVAC system during the field audit, it was noticed that most of the humidifiers were operating at 100% load as can be seen in the below example while the outdoor ambient conditions (fresh air) had an RH larger than 35%. The reason is that the setpoint was set at 45% on the return air humidity levels.

The impact of this unnecessary humidification are two key points:

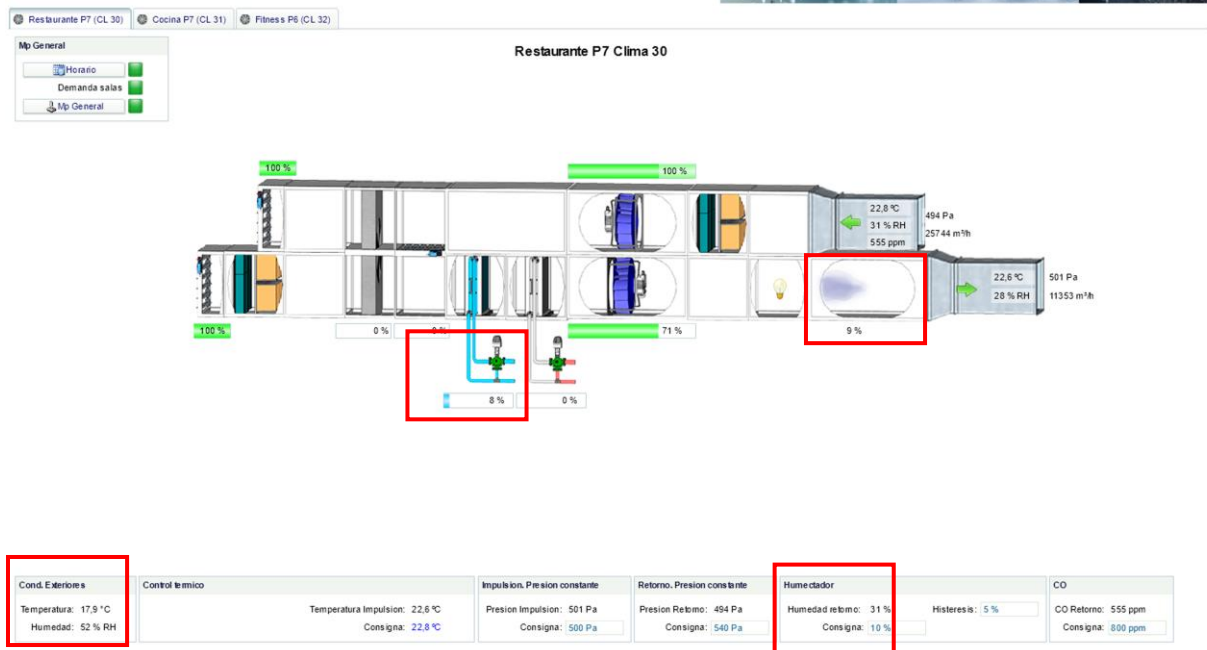
- i. The electrical demand/consumption of humidifiers are full
- ii. The humidification process increases back the supplied temperature which pushes the cooling coil to open to compensate back to bring down the temperature to the needed setpoint – while external temperatures were below 18°C. This is therefore pushing the chilled water/heat pump operations to increase unnecessarily too.

In both examples below: RH setpoint was more than 40%, Outdoor temperature was below 18°C, Humidifiers running at 100% and cooling coil open at 16% and 100% respectively.

Figure 32: Various AHUs operation



In the below example, Ambient temperature was 18°C with a RH of 52%, RH setpoint was 10% (i.e: we were testing it) and still the humidifier didn't stop completely and kept running at 9% despite the RH values on the return air was 31%. Note also the cooling coil opening, even if it was just 8% knowing that the supply temperature setpoint is 22.8°C and the ambient air was 17.9°C!



The recommended actions are as follows:

- Reduce all RH setpoints from their current levels to 20% in all AHUs serving BOH areas
- Reduce all RH setpoints from their current levels to 25% in all AHUs serving other areas except the fresh air intake in guest rooms
- Reduce all RH setpoints from their current levels to 30% in all AHUs serving the Fresh Air Handling Units serving guest rooms
- Ensure that if a RH level reaches the setpoint , the humidifier should be totally shut down (0% load)
- Review all sensors across all the process to ensure readings that are leading the automation are correct.

In principle, low humidity only arises in summer months and in the remaining 8 months , humidifiers should be seen operating at maximum 20% of the time.

The savings to be attained are very substantial:

- As seen in the table in the following page, humidifiers have a total installed Power of 266KW which is larger than the combined Fans of all AHUs! Implementing the strategy will reduce the related demand eaaily by 75% during 8 out of 12 months of the year. This will also help reduce the total power drawn from the utility – in line with the first discussed measure of reducing subscribed power in the 6 time of use slots
- Electricity savings would be substantial and will be calculated in more accuracy in the final energy audit report. However, **they can be safely estimated to top the 700,000KWh annually equivalent to €192,500** based on the latest tariff of €0.275/KWh

- iii. It is important to note that savings will be at the levels of humidifiers as well as chillers/heat pumps and their set of pumps as the cooling demand will drop.

Table 13: AHUs and their humidifiers power rating

AHU	KW Humidifier
7-CL30	15
6-CL 29	15
6-CL13	25
5-CL12	37
3-CL4	7.52
3-CL6A	3.7
3-CL6B	3.7
3-CL8A	15
3-CL8B	15
3-CL11	15
3-CL14	30
3-CL15	22
3-CL36	3.7
2-CL09	15
2-CL28	6
2-CL07	7.52
S1-CL01	15
S1-CL37	15
TOTAL (KW)	266.14

EEM - 04		
HVAC Control: Humidifiers' operation		
ENERGY SAVINGS		
ELECTRICITY	GAS	CO ₂
KWh	m3	Kg
999,519	0	298,640
14.0%	0.0%	14.0%
ECONOMIC PERFORMANCE		
Annual Cost Savings [€]		189,652 €
Annual Share of Cost Savings [%]		11.4%
Implementation Cost [€]		0 €

5.5. EEM-05: HVAC Control: Review all pressure setpoints → Control of VFDs

All AHUs' Fans are equipped with VFDs that are operated on a PID loop according to the pressure of the supplied and/or extracted air v/s a setpoint that is set for both fans as per the below example. The logic behind each pressure setpoint is not clear and doesn't seem logical from unit to unit. When an AHU is of a VAV type, the pressure is set based on the designed maximum airflow versus the number of VAV boxes open, when it is a CAV, the pressure is not related to anything but to control the fan itself according to the volume of air.

Figure 33: AHU example of pressure setpoints



The key efficiency actions are the following:

- Review all AHUs with their specific type, pressure setpoints and area serviced. Then modify the setpoints to allow the fans to be modulated lower as per the below two possibilities:
- Review all the PID loops of the VFD of VAV AHUs ensuring an aggressive programming. Then test the impact of the pressure on the VFD feedback.
- Alternatively, implement a scheduled based VFD operation with a lower speed in off peak seasons (winter) or specially at nighttime in all AHUs (CAVs and VAVs)

In short, VFDs are not meant to remain at a fixed speed and as much as you are able to lower the speed as you are increasing the energy savings. Similarly, pressure setpoints can also be adjusted proactively according on a time, seasons or occupancy based.

The potential savings is minimum 20% on the electricity consumption of the Fans of AHUs on the basis that a 10% reduction of speed will lead to a 27% reduction in its electricity consumption.

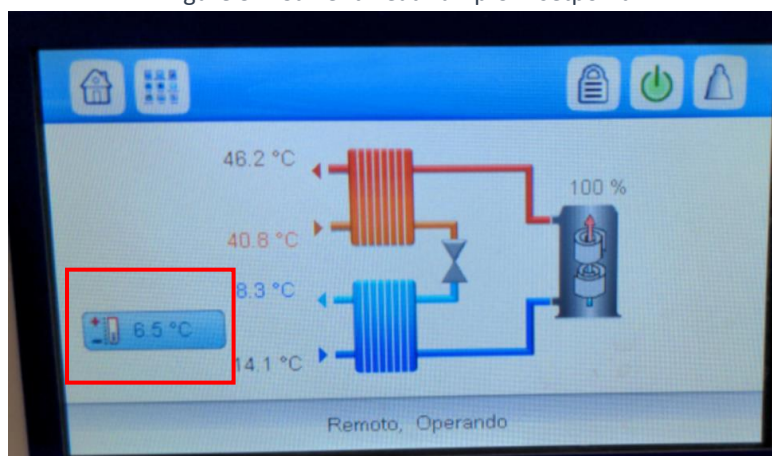
EEM - 05		
HVAC Control: Review all pressure setpoints Control of VFDs		
ENERGY SAVINGS		
ELECTRICITY	GAS	CO ₂
KWh	m3	Kg
215,028	0	64,247
3.0%	0.0%	3.0%
ECONOMIC PERFORMANCE		
Annual Cost Savings [€]		40,800 €
Annual Share of Cost Savings [%]		2.4%
Implementation Cost [€]		0 €

5.6. EEM-06: Chillers/Heat Pumps: Chilled Water Temperature Reset

Generally, Chilled-water reset adjusts the chilled-water set point to improve the efficiency of the chiller and reducing its energy consumption. Usually, a chilled-water-reset strategy raises the set-point temperature when the building load is at less-than-design conditions. Producing warmer chilled water lessens the burden on the compressor, which means that the chiller consumes less energy.

In the FSM chilled water outlet temperature was traditionally fixed, usually somewhere around 6.5°C. It is advisable now to apply a chilled water reset strategy the temperature of the chilled water to be increased during part load conditions which as indicated reduces the amount of work done by the chillers compressor.

Figure 34: Current Heat Pump CW setpoint



The increase in chilled water temperature results in an increase in evaporator pressure of the refrigerant. The compressor takes the refrigerant at the evaporator pressure then compresses it and increases its pressure to the condensers pressure, so the higher the evaporator pressure the less work the compressor needs to do to increase that pressure.

In peak winter, the temperature can be also increased further during nighttime (for example to 11°C and then dropped back to 9.5°C after 8hAM). As a rule-of-thumb, each 0.5°C increase in the chilled-water temperature reduces the energy consumption of the chiller by an amount from 1 percent to 1.5 percent.

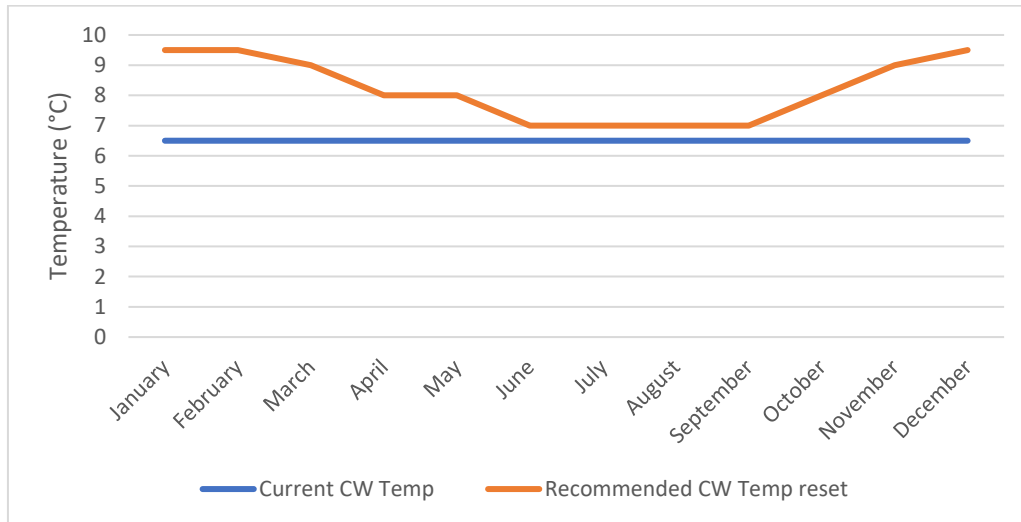
The recommended strategies to be implemented are as follows:

- 8°C during the months of October, April and May
- 7°C to 7.5°C during the months of May to September.
- 9°C during the months of March and November
- 9.5°C during the months of December to February

Those settings can be proactively changed throughout the year as the idea is to push the chilled water network to the highest possible leaving temperature while maintaining guests

and employees satisfaction by meeting the demand load. With the very long and large chilled water piping infrastructure and the related volume of water that is circulating, any difference in temperature setting will have a big impact on the consumption.

Figure 35: Proposed Monthly Reset Temperature



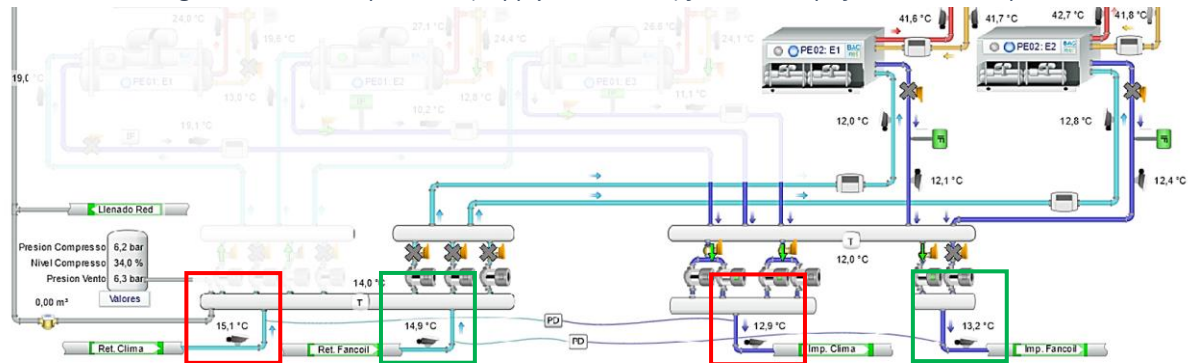
The **total electricity savings to be attained would be minimum 3% from the total chillers'** electricity consumption.

EEM - 06		
Chillers/Heat Pumps: Chilled Water Temperature Reset		
ENERGY SAVINGS		
ELECTRICITY	GAS	CO ₂
KWh	m3	Kg
27,968	0	8,356
0.4%	0.0%	0.4%
ECONOMIC PERFORMANCE		
Annual Cost Savings [€]		5,307 €
Annual Share of Cost Savings [%]		0.3%
Implementation Cost [€]		0 €

5.7. EEM-07: Chilled Water: Proactive ΔP Setpoint reset at Secondary Chilled Water Pumps

During the field audit it was seen that the Delta Temperature (ΔT) at both chilled water loops (FCUs and AHUs) is very low with 2.1°C and 1.7°C respectively.

Figure 36: CW Temperature (Supply and Return) for both loops from Heat Pumps



In fact, both chillers and heat pumps operations (summer and winter) operate at a fixed ΔP setpoint (differential pressure across the time).

An unnecessarily high ΔP setpoint will push the pumps to operate at higher speed with more CW flow and lower ΔT pushing chillers to run at lower efficiencies too.

The importance of the Delta P is at two levels:

- It enhances the operation of the secondary pumps further by reducing the chilled water flow demand.
- It enhances the Delta Temperature (ΔT) as an over pumping will reduce Delta T and will increase the chillers' capacity/consumption. The below graph showcases ΔT over 6 months prior to the field audit. It is clearly seen that in peak summer it tops an average of 4°C which can be considered as mildly satisfactory but it goes down to 2°C in November and eventually during winter times.

The action to be taken is simply to change proactively the Delta P setpoint from its current levels down to the lowest possible value. This can be done on a 'trial and error' basis by going down and analyzing the overall cooling performance across the building. The current levels may be close to the final ones during peak summer seasons but from these 4 peak months levels could go gradually down to reach the lowest possible threshold in January/February.

The proposed actions are as follows (to be implemented on secondary and all tertiary pumps):

- Set the lowest possible ΔP in peak summer months by starting to reduce the existing one by notches of 0.1 bars and analyzing the while dynamics of the chilled water system and meeting the building cooling demand. This will allow you to set the maximum setpoint of ΔP

- ii. Repeat the above exercise in peak winter month to find the lowest possible threshold
- iii. Have the team proactively changing the setpoint in between the above two thresholds across all months by ensuring it is always to the minimum possible. Parameters affecting the cooling demand such as the ambient temperature, humidity and occupancy should be tracked along with the above figures.

The impact of properly and proactively modifying the ΔP across the seasons and conditions will be at the level of both the chiller's capacity and reducing the CW secondary pumps' speed and electricity consumption.

As we are currently in peak winter, it is a perfect time to test the lowest threshold possible for the ΔP which could be easily 30% lower from the current levels.

Overall savings are from the electricity reductions of all secondary chilled water pumps – estimated an average of 15% annually- as well as on the chillers due to the optimized ΔT estimated at about 3% annually.

EEM - 07		
Chilled Water: Proactive ΔP Setpoint reset at Secondary Chilled Water Pumps		
ENERGY SAVINGS		
ELECTRICITY	GAS	CO ₂
KWh	m3	Kg
49,952	0	14,925
0.7%	0.0%	0.7%
ECONOMIC PERFORMANCE		
Annual Cost Savings [€]		9,478 €
Annual Share of Cost Savings [%]		0.6%
Implementation Cost [€]		0 €

6. Residences section

6.1. General Description

FSM property includes 22 private residences that have their technical infrastructure totally separated from the hotel side.

The key technical features are as follows:

- i. Chilled Water/Space cooling : The residences' space cooling is done through two air cooled Heat Pump Chillers , model Carrier 30RBP-260 (2650KW capacity each) with R410 refrigerant. The units have 2 circuits and run with VFD condenser fans. The chilled water system includes three primary pumps and two sets of Secondary Chilled Water Pumps servicing the FCUs and AHUs respectively. At the time of the field audit, chilled water was produced at 8°C

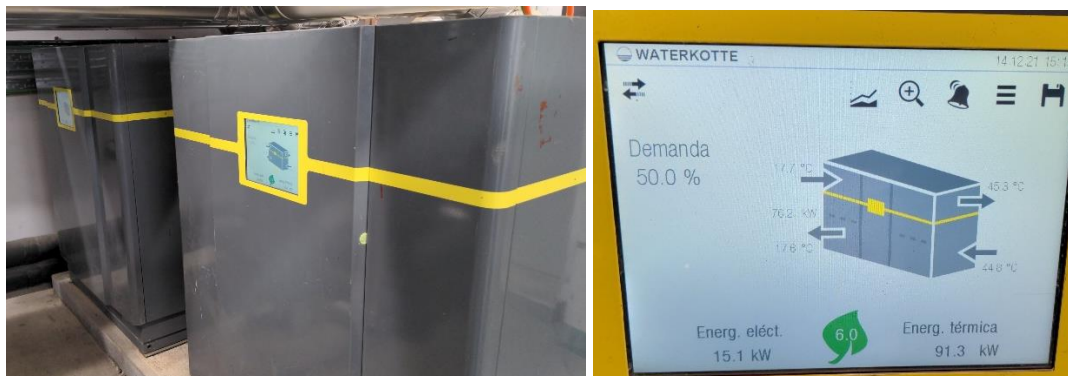
Figure 37: Carrier 30 RBP-260 Air cooled chillers (Left); Secondary Chilled Water Pumps (Right)



- ii. Thermal Energy: The thermal systems of the residences include two systems: A Ground Source Heat Pump and a gas driven boiler.

The Ground Source Heat Pump (GSHP) system consists of two Heat pumps units model Waterkotte Ecotouch 5145 with a 111KW to 146KW capacity of heating.

Figure 38: GSHP – 2 units and operation during field audit





Their condenser side is connected to two main ground source collectors each having three boreholes. The GSHP primary role is to provide heating for the floor heating systems across all residences but it has also one heat exchanger connected to the hot water system. During the field audit the COP was seen to be varying between 4.5 and 6 which is very satisfactory. Temperature produced was between 45°C and 47.4°C.

The second part of the thermal energy consists of the boiler house. The latter includes two gas driven boilers Viessman Vitrocrossal 200 – 232KW capacity each.

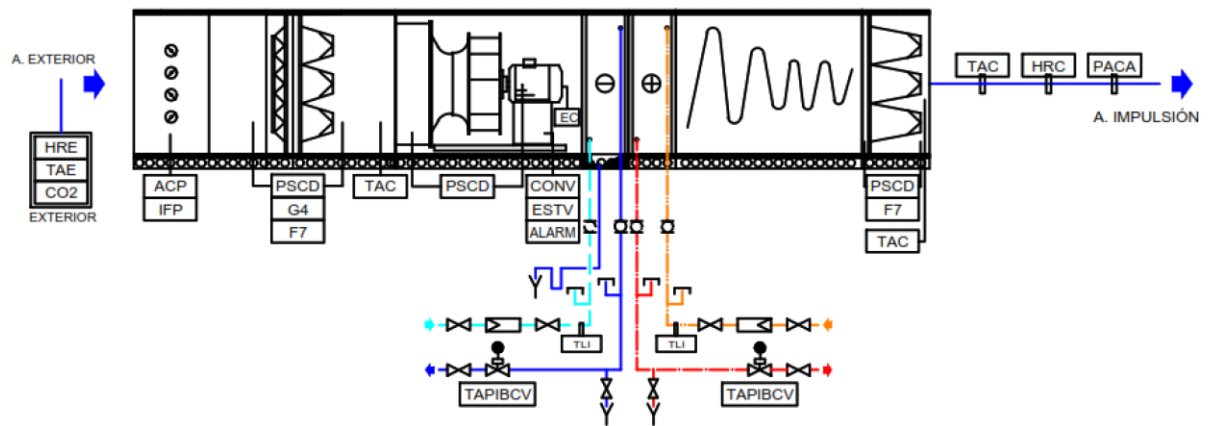
There are 4 2000 liters calorifiers in the plant, 3 are serviced from the boilers with a temperature set at 64°C while the fourth one is connected to the GSHP system.

Figure 39: Gas Driven Boilers (Left) – Calorifiers (Right)



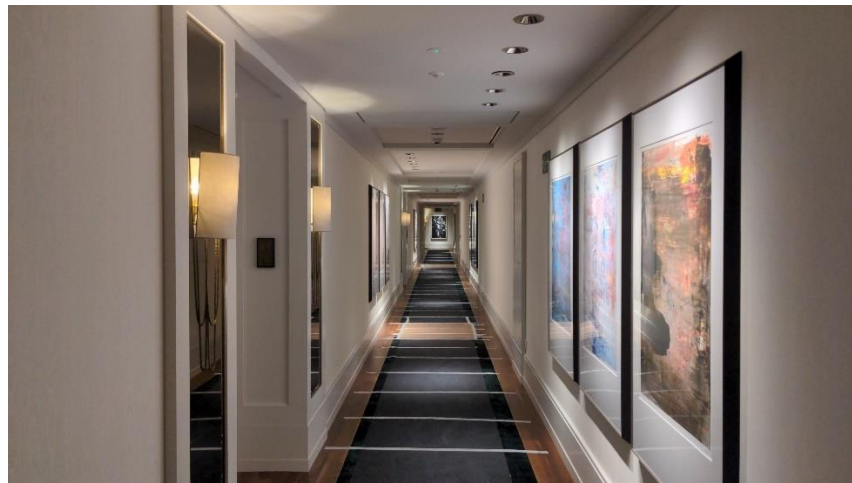
- iii. Ventilation : Public/common spaces ventilations include a primary AHUs as well as a number of extractor fans . Below is the key schematic of the Fresh air AHU that serves the whole areas.

Figure 40: Primary Fresh AHU for the Residences



- iv. Lighting is solely composed of LED lights and there are only small common area parts including corridors , lobby and a lounge

Figure 41: Typical Residential floor's corridor



- v. Metering : Apart the shared/common infrastructure, residences have submetering for electricity , chilled water , hot water and water usage . They are the basis of the monthly charging .

6.2. Recommended Energy Efficiency Measures

Just like in the hotel part, the EEMs foreseen for the residences are all in the no cost category. Overall the chosen systems and equipment have high efficiency and it is the appropriate operations that will enhance further the energy efficiency.

The key EEMs are the following actions :

i. Chilled Water Optimization :

This would include the application of **EEM06** and **EEM07** previously discussed for the hotel but in the residential's chillers systems. Mainly, apply the following strategies:

- Reset Chilled Water Temperature , increasing temperature as high as 10°C in winter and maintaining it at 8°C in summer months.
- Proactive change of DP on the secondary chilled water pumps. Up to 50% decrease is possible in peak winter months (i.e: if the present DP is 2 bars, reduce it to 1 bar) and up to 15% in summer months, This would be obviously driven by occupancy and climatic conditions.

ii. Operation of the FAHU (Primary AHU) and other fans:

- Ensure that whenever ambient temperature is between 15C and 22C the primary AHU has the cooling/heating coil disabled in a way to distribute the air directly as is but filtered, Any needed cooling/heating will be done locally in the residences using their FCUs. This will optimize the 'Free air ' and 'Free Cooling' concepts.
- If the occupancy is low, the FAHU can be stopped 25% of the time during the day easily , specially that fresh air is taken through doors and windows while occupancy in various spaces is generally low.
- All fans can be duty cycled depending on occupancy and hours of the day (stopping them for example 30mn every 2 hours is possible during daytime if planned according to known utilization of spaces and occupancy.
- When ambient temperatures are below the levels seen in the first point and then the cooling or heating coils are to be operated, ensure temperature setpoint are set at the lowest possible in winter (20°C) and highest in summer (22°C or 23°C). Any additional cooling or heating will be done locally.

iii. Geothermal system in summer months

The GSHP primary usage is for the floor heating across all residences. However, during at least the 6 to 8 months when there is no need for floor heating, the full capacity of the heat pump units should be used to preheat fully the make up water of the domestic hot water system. They will increase the temperature up to 47°C which will ensure gas boilers will work the minimum. With a COP of up to 6, the GSHP are a very efficient operation.